
Provable Learning of Linear Energy Models over Function Spaces

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1 Introduction

Abstract

Transformers with billions of parameters have achieved remarkable success across various domains, such as language modeling and computer vision. Existing works analyze the empirically observed phenomena, such as training dynamics, in-context learning (ICL) generalization, depth/width effects, scaling laws, only in isolation and in very restricted settings (single-layer, linear regression, fixed width), leaving open whether they are manifestations of a single underlying mechanism. In this paper, we show how a Transformer-like Energy-Based Model (TEBM) can provably learn a Gibbs distribution over a function space $f \propto \exp(-E_\theta(f))$ from noisy observations $f_j(x_i) + \epsilon$. We consider linear families of energy functionals and show how in the mean-field regime, where the number of heads $\rightarrow \infty$, training a TEBM recovers the parameter θ with provable approximation and generalization guarantees. Our findings prove the convergence of the unsupervised training dynamics of TEBMs, the emergence of scaling laws in the considered setting, and the necessity of the regularization of θ for the generalization properties of the ICL. We evaluate our architecture on synthetic and real-world benchmarks to support our theoretical findings.

This paper makes the following contributions:

- **Learning distributions over functions.** We formulate the problem of learning a distribution ν over a function space $\mathcal{F}$ from noisy observations as a regularized optimization problem. We model ν as an exponential family with sufficient statistic $\Phi: \mathcal{F} \rightarrow \mathbb{R}^d$, so that $p(f) \propto \exp(-\langle \theta, \Phi(f) \rangle)$, and state generalization guarantees for several (semi-)parametric choices of Φ .
- **Connection to in-context learning.** We draw a connection of this problem to the in-context learning in Transformer-like Energy-Based Models, discuss how it can explain the convergence to the optimal solution, the empirically observed scaling laws, the role of the positional encodings and the importance of the regularization for the generalization properties.
- **Architectural implications and empirical validation.** Guided by the optimization problem, we propose a modification of the Transformer architecture that enforces the structural assumptions of our framework. Experiments on both synthetic and real-world benchmarks demonstrate improvements in size-generalization, interpretability, and computational efficiency.

34 2 Related Work

35 Several works have showed that a fixed Transformer can implemet a gradient descent for a simple
36 class of functions such as linear regression, Bayesian inference, etc.

37 3 Problem Setup

38 We are given a manifold $\mathcal{M}$ equipped with a probability measure μ , a function space $\mathcal{F}$ of functions
39 $f: \mathcal{M} \rightarrow \mathbb{R}$, endowed with an *unknown* probability measure ν over $\mathcal{F}$. We observe a dataset
40 $\mathcal{D} = \{D^{(1)}, \dots, D^{(m)}\}$ consisting of m observations $D^{(j)}$ of functions $f_1, \dots, f_m \stackrel{\text{i.i.d.}}{\sim} \nu(\mathcal{F})$, each
41 evaluated at n points drawn from μ :

$$x_1, \dots, x_n \stackrel{\text{i.i.d.}}{\sim} \mu(\mathcal{M}), \quad \epsilon_{j,i} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2),$$

$$\mathcal{D} = \left\{ \left\{ (x_i, f_j(x_i) + \epsilon_{j,i}) \right\}_{i=1}^n \right\}_{j=1}^m.$$

42 Our goal is to learn the measure ν over $\mathcal{F}$. Given ν , predicting a new function $f \sim \nu$ from noisy
43 observations $\{(x_i, y_i)\}_{i=1}^n$, where $y_i = f(x_i) + \epsilon_i$ and $\epsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$, reduces to MAP estimation:

$$\hat{f}_m = \arg \min_{f \in \mathcal{F}} \sum_{i=1}^n |y_i - f(x_i)|^2 - \log \nu(f).$$

44 **Connection to ICL in Transformers.** The setup described above is closely related to in-context
45 learning in Transformers. Given a prompt $(x_1, y_1), \dots, (x_n, y_n)$, where x_i are positional encodings
46 and y_i are tokens, the task is to predict the next token at position x_{n+1} . Each prompt can be viewed
47 as a noisy observation of a latent function $f \sim \nu(\mathcal{F})$, and distinct prompts correspond to distinct
48 draws from $\nu(\mathcal{F})$. The Transformer thus implicitly learns the shared structure across functions, that
49 is, the measure ν itself, and uses it to generalize within a new context.

50 4 Learning Energy Functionals In Theory And Practice

51 In this section, we derive an optimization problem to address the goal described in Section 3.

52 **Assumption 4.1.** The function domain $\mathcal{M}$ is a smooth Riemannian manifold embedded in a Euclidean
53 space $\mathcal{M} \subset \mathbb{R}^d$ equipped with a Riemannian metric $\langle \cdot, \cdot \rangle$ and a probability measure μ over it.

54 **Assumption 4.2** (Gibbs distribution). We assume that the probability measure ν over $\mathcal{F}$ is a Gibbs
55 distribution induced by a convex energy functional $E_\theta: \mathcal{F} \rightarrow \mathbb{R}^+$ and a reference measure (**TODO:**
56 **define**) ν_0 on $\mathcal{F}$:

$$\nu(f) = \frac{1}{Z_\theta} \exp(-E_\theta(f))$$

57 where $Z_\theta = \int_{f \in \mathcal{F}} \exp(-E_\theta(f)) \nu_0(df)$ is a normalizing constant and E_θ is a family of functionals
58 parametrized by $\theta \in \Theta$.

59 4.1 Function Extrapolation

60 First, let us see how knowing the energy functional enables function extrapolation. Suppose we
61 observe $D = \{(x_i, y_i)\}_{i=1}^n$ with $y_i = f(x_i) + \epsilon_i$, $\epsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$, and wish to predict $y_{m+1} =$
62 $f(x_{m+1})$ at a new point x_{m+1} . Given the Gibbs prior $p(f) \propto \exp(-E(f))$, we recover f via MAP
63 estimation:

$$\hat{f}_n = \arg \max_{f \in \mathcal{F}} p(D | f) p(f) = \operatorname{argmin}_{f \in \mathcal{F}} \frac{1}{\sigma^2} \sum_{i=1}^n |y_i - f(x_i)|^2 + E(f).$$

64 This estimation will have a generalization guarantee of extrapolating f to a new point $x_{n+1} \sim \mu$ via
65 the following theorem:

66 **Theorem 4.3.** *[TODO: add theorem]*

67 For example, assuming that our function is s -smooth on $\mathcal{M} \subset \mathbb{R}^d$ then Theorem 4.3 recovers the
 68 standard rate $O(n^{-2s/(2s+d)})$ of the minimax rate of estimation of a smooth function from n noisy
 69 observations.

70 4.2 Likelihood

71 As we mentioned, the family of the energy functionals E_θ depends on some parameter θ . Hence,
 72 our goal is to learn the parameter θ by maximizing the likelihood of the observed data. First, let
 73 us consider a simpler case, when we are given full functions $\mathcal{D} = \{f_1, \dots, f_m\}$, not just empirical
 74 samples $f_i(x_j) + \epsilon$ and we want to learn the parameter θ by maximizing the likelihood of the observed
 75 data.

$$\min_{\theta} -\log p_{\theta}(\mathcal{D}) = \frac{1}{m} \sum_{i=1}^m E_{\theta}(f_i) + \log Z(\theta)$$

76 where $Z(\theta) = \int_{f \in \mathcal{F}} \exp(-E_{\theta}(f)) \nu(df)$ is a normalizing constant, ν base reference measure on $\mathcal{F}$.

77 Now, back to the original problem, where each function f is given by its values: $D = \{(x_i, y_i)\}_{i=1}^n$
 78 such that $y_i = f(x_i) + \epsilon_i$, $\epsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$. So given an observation $D^{(i)} = \{(x_j, y_j)\}_{j=1}^n$ our
 79 optimization problem turns into

$$\min_{\theta} \mathcal{L}(\theta) = -\frac{1}{m} \sum_{i=1}^m \log p_{\theta}(D^{(i)}) + \log Z(\theta) \quad (1)$$

$$\text{where } \log p_{\theta}(D^{(i)}) = \log \int_{f \in \mathcal{F}} \exp(-\log p(D^{(i)} | f) - E_{\theta}(f)) \nu(df)$$

80 Unfortunately, in practice each term $\log p_{\theta}(D^{(i)})$ is intractable. However, we can approximate it by
 81 a bilevel optimization problem.

82 4.3 Bilevel Optimization Via Masking

83 In this section, we discuss how to approximate our intractable convex optimization problem (1) by a
 84 bi-level non-convex optimization problem via masking approach. Fix a parameter θ and a dataset
 85 $D = \{(x_j, y_j)\}_{j=1}^n$. Define $L_{\theta}(f, D) := \ell(D, f) + \lambda E_{\theta}(f)$, where $\ell(D, f)$ denotes the empirical
 86 negative log-likelihood and E_{θ} is an energy functional.

Note, for a fixed θ the minimizing $L_{\theta}(f, D)$ with respect to f is unique: $\hat{f}_{\theta} = \arg \min_{f \in \mathcal{F}} L_{\theta}(f, D)$
 Moreover, the posterior distribution $p_{\theta}(f | D) \propto \exp(-L_{\theta}(f, D))$ concentrates in probability
 around $\hat{f}_{\theta}$ as either $n \rightarrow \infty$ or $\sigma^2 \rightarrow 0$. In particular, for any neighborhood U of $\hat{f}_{\theta}$ in $\mathcal{F}$,
 $p_{\theta}(U | D) \rightarrow 1$. Using a second-order Taylor expansion,

$$E_{\theta}(f) = E_{\theta}(\hat{f}_{\theta}) + \frac{1}{2} \langle f - \hat{f}_{\theta}, H_{\theta}(f - \hat{f}_{\theta}) \rangle + o(\|f - \hat{f}_{\theta}\|_X^2).$$

87 where Hessian operator $H_{\theta} := \nabla^2 E_{\theta}(\hat{f}_{\theta})$ is the second-order derivative of the energy functional E_{θ} ,
 88 we can approximate the intractable integral as:

$$\int_X \exp(-\log p(D | f) - E_{\theta}(f)) \nu(df) \approx \exp(-\log p(D | \hat{f}_{\theta}) - E_{\theta}(\hat{f}_{\theta})) (\det H_{\theta})^{-1/2}. \quad (2)$$

89 Substituting the Laplace approximation (2) into the negative log marginal likelihood yields

$$-\log p_{\theta}(D) \approx \log p(D | \hat{f}_{\theta}) + E_{\theta}(\hat{f}_{\theta}) + \frac{1}{2} \log \det H_{\theta} + \log Z(\theta) \quad (3)$$

90 up to additive constants independent of θ . Using this approximation, we can turn the intractable
 91 convex optimization problem (1) into a tractable bilevel non-convex surrogate of it. Given the

92 collection of datasets $\mathcal{D} = \left\{ \left(D^{(i)}, D_{\text{mask}}^{(i)} \right) \right\}_{i=1}^m$. Substituting this into Equation (1) we have

$$\min_{\theta \in \Theta} \mathcal{L}(\theta) = \frac{1}{m} \sum_{i=1}^m \log p \left(D_{\text{mask}}^{(i)} \mid \hat{f}_{\theta}^{(i)} \right) + E_{\theta} \left(\hat{f}_{\theta}^{(i)} \right) + \frac{1}{2} \log \det H_{\theta}^{(i)} - \log Z(\theta) \quad (\text{outer problem})$$

$$\text{where } \hat{f}_{\theta}^{(i)} = \operatorname{argmin}_{f \in \mathcal{F}} \log p \left(D^{(i)} \mid f \right) + E_{\theta}(f) \quad (\text{inner problem}) \quad (4)$$

93 The inner problem has parameter θ fixed and find the function $\hat{f}_{\theta}^{(i)}$ that minimizes the loss + energy
 94 functional. The outer problem then finds the parameter θ yields the best prediction of $\hat{f}_{\theta}^{(i)}$ on the
 95 masked dataset $D_{\text{masked}}^{(i)}$. Both problems are convex on their own but not jointly convex. However, we
 96 can solve it by alternating gradient descent with a following guarantee:

97 **Theorem 4.4.** *[TODO: add theorem]*

98 **The Inner Problem.** For a dataset $D = \{(x_i, y_i)\}$ we have

$$\hat{f}_{\theta} = \operatorname{argmin}_f \log p(D \mid f) + E_{\theta}(f) = \operatorname{argmin}_f \|f(x_i) - y_i\|_2^2 + \langle \theta, \Phi(f) \rangle$$

99 Note, that due to convexity, this can be solved by subgradient descent on f with a step size $\eta > 0$:

$$f^{(t+1)} = f^{(t)} - \eta \left[2 \sum_{i=1}^n (f^{(t)}(x_i) - y_i) \delta_{x_i} + \nabla_f E_{\theta}(f^{(t)}) \right]$$

100 where $\nabla_f E_{\theta}(f^{(t)})$ is a (sub)gradient of the energy functional.

101 **The Outer Problem.**

102 5 Families Of Energy Functionals

103 Here we look closer at specific linear families of energy functionals $E_{\theta} = \langle \theta, \Phi(f) \rangle$ defined by the
 104 statistic extractor Φ . We discuss what inductive bias each choice of Φ induces, what generalization
 105 properties it can achieve and how a transformer forward pass can be similar to a (sub)gradient descent
 106 for the inner problem, and transformer training can be seen as a gradient descent on the outer problem.

107 **Notation** Let $\mathcal{Q} \subset \mathbf{S}_+^d$ be *[TODO: define]*. Given a PSD matrix Q , let Δ_Q denotes the
 108 anisotropic Laplacian operator: $\Delta_Q f := \nabla \cdot (Q \nabla f)$. Total variation norm is defined as
 109 $\|g\|_{\mathcal{M}} := \int_{x \in \mathcal{M}} \|g(x)\|_{\mathcal{M}} dx$ where g *[TODO: can be a Radon measure.]*

110 **Anisotropic Sobolev regularization.** Let θ be a measure over $\mathcal{Q}$. Let statistic $\Phi : H^1(M) \rightarrow \mathbb{R}^k$
 111 be defined as $\Phi(f)(Q) := \int_{x \in \mathcal{M}} \|\nabla f(x)\|_Q^2 dx$. Then:

$$E_{\theta}(f) = \int_{Q \in \mathcal{Q}} \Phi(f)(Q) d\theta(Q) \quad \nabla_f E_{\theta}(f) = \int_{Q \in \mathcal{Q}} \Delta_Q f d\theta(Q)$$

112 **Anisotropic second-order TV** Define statistic $\Phi : X \rightarrow \mathcal{C}(\mathcal{S}_+, \mathbb{R}_+)$ as $\Phi(f)(Q) := \|\Delta_Q f\|_{\mathcal{M}}$.
 113 Then:

$$E_{\theta}(f) = \int_{\mathcal{Q}} \|\Delta_Q f\|_{\mathcal{M}} d\theta(Q) \quad \nabla_f E_{\theta}(f) = \int_{\mathcal{Q}} \Delta_Q \operatorname{sign}(\Delta_Q f) d\theta(Q)$$

114 This promotes sparse, geometry-aligned curvature and induces a Finsler total variation.

115 **Spectral energy features.** Let $\{\psi_k\}$ be eigenfunctions of a Laplacian on M . Define $\Phi : L^2(M) \rightarrow$
 116 ℓ^1 , $\Phi_k(f) := |\langle f, \psi_k \rangle|^2$.

117 The corresponding regularizer weights energy by frequency and induces a spectral bias.

118 6 Connection to Transformers And Its Implications

119 In most of the examples above, the key step of implementing a (sub)gradient descent for the inner
 120 problem is to compute the discrete (anisotropic) Laplacian Δ_Q for some PSD matrix Q . The original
 121 Laplacian Δ_Q can be approximated by its discrete version $\hat{\Delta}_Q \in \mathbb{R}^{n \times n}$ computed on a set of points
 122 $x_1, \dots, x_n \in \mathcal{M}$:

$$\hat{\Delta}_Q = I - D^{-1/2} W D^{-1/2} \quad \text{where}$$

$$W_{ij} = \exp\left(\frac{-\langle x_i - x_j, \Pi_{x_i} Q^\dagger \Pi_{x_i}(x_i - x_j) \rangle}{\epsilon}\right) \quad \text{and} \quad D_{ii} = \sum_{j=1}^n W_{ij}.$$

123 Even though the attention mechanism is not exactly the same as the discrete Laplacian, it is similar to
 124 it in a few ways. The attention mechanism is a way to compute the weighted average The discrete
 125 Laplacian is a way to compute the weighted average of the features of the points in the neighborhood
 126 of a given point. Consider a transformer-like architecture where the softmax attention is replaced
 127 with 2 different types of heads per layer:

- 128 • 1 **linear** head: $y_{t+1} = (I)$
- 129 • H **diffusion** heads each parametrized by some $Q_h \in \mathcal{Q}$ and $\theta_h \in \mathbb{R}^+$: $\bar{y}^{(t+1)} = I -$
 130 $\eta \theta_h \Delta_{Q_h} \bar{y}^{(t)}.$

131 Then the forward pass of

